

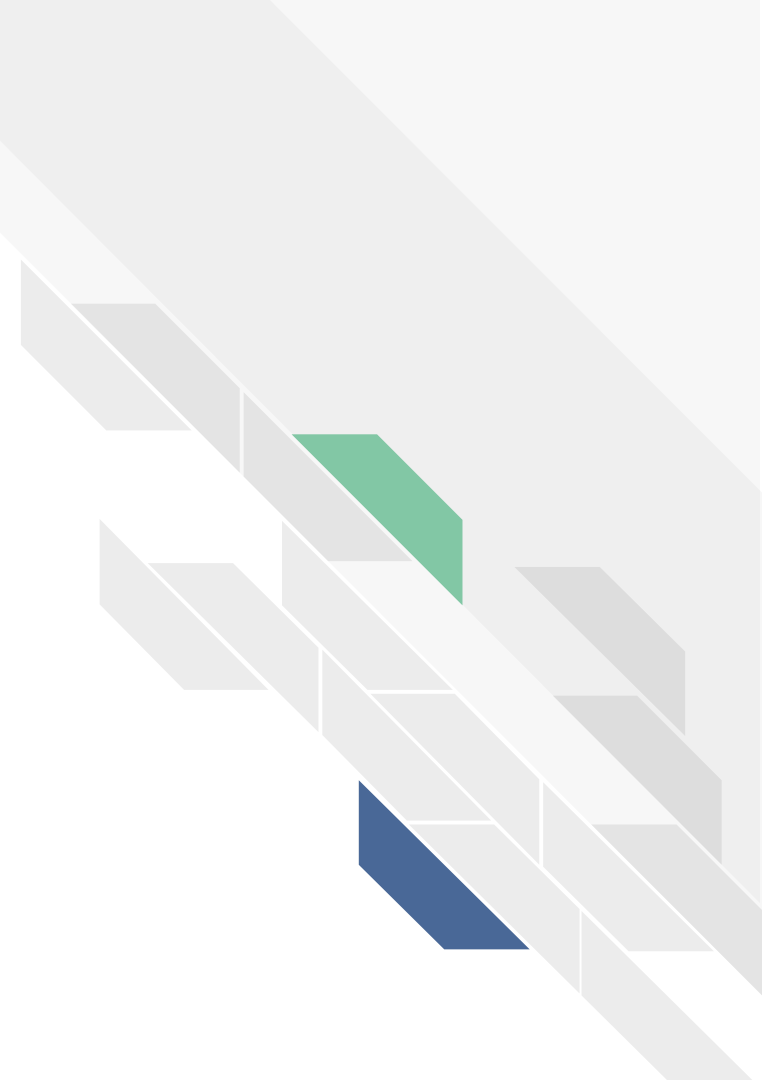


Initial Works Like Presentation

Ortho Team

Bianca, Braxton, Jonathan, Li

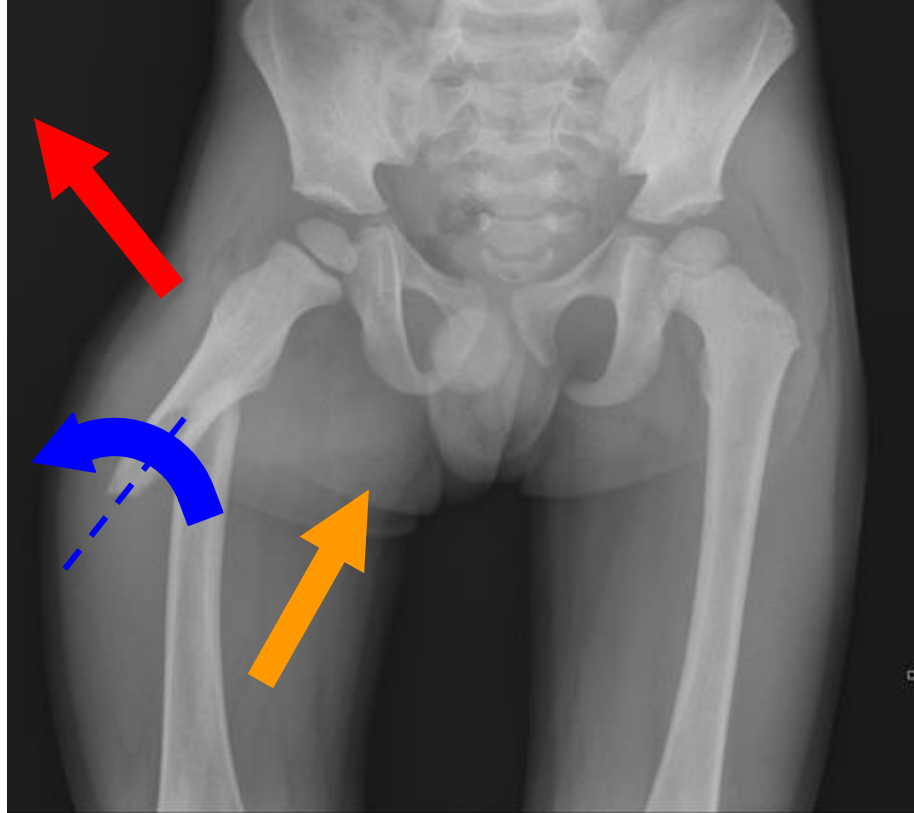
Problem and Need Statement



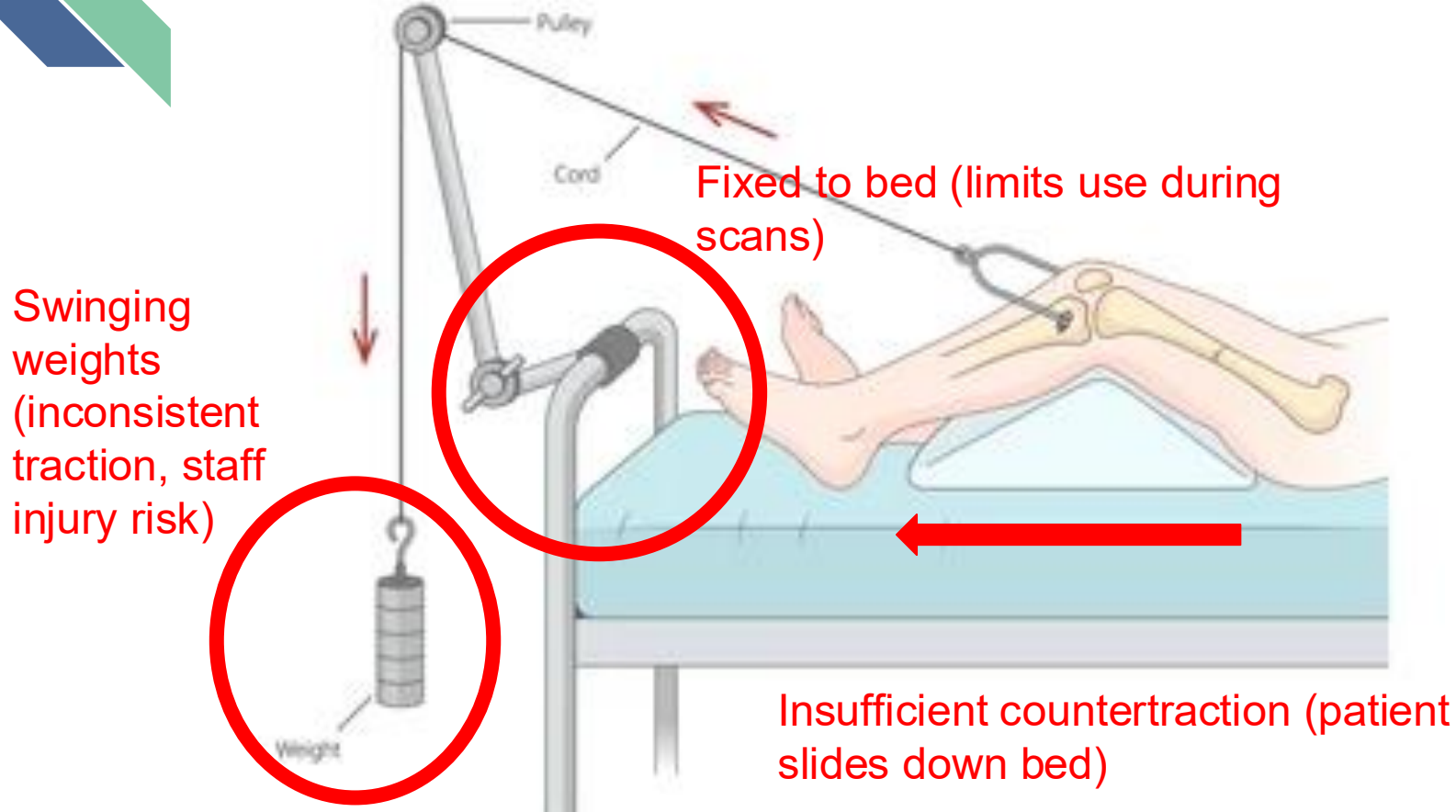
Problem Recap

Fracture fragments are displaced due to muscle contractions.

To counteract this displacement, skeletal traction is a damage control method that realigns the femur by pulling the femur to length with weights.



Issues with Current Standard of Care





Need Statement

Orthopedic surgeons need a way to apply skeletal traction to adult patients with femoral shaft fractures that maintains correct femur length throughout treatment despite changes in patient position to prevent femur displacement that can cause further damage.

Current Technology Limitations



Expensive!

Either portable OR
sufficient traction

No indication of how
much force is
applied





Device Requirements

	User Need		User Need
1.	Device shall apply traction to femur fracture per current treatment requirements	6.	The pin must align with current traction pin use and safety standards
2.	The device shall maintain the set length and position of femur throughout treatment	7.	Device shall allow for timely evaluation, monitoring, and cleaning of pin site
3.	User must be able to know forces applied	8.	Device pin shall be biocompatible
4.	Device shall be radiolucent for X-rays and CT scans	9.	Device pin shall be sterilizable
5.	The device must be compatible with use of most hospital bed, including X-ray and CT beds	10.	Device packaging and labeling shall be consistent with FDA standards
		11.	Device shall be composed of reusable components



User Need	Design Input
Device shall apply traction to femur fracture per current treatment requirements	Device must be able to apply 5-40 lbs (10-20% of patient's body weight) of traction axially
	Apply in increments of 2.5 lbs
The device shall maintain the set length and position of femur throughout treatment	Force applied by the device does not deviate more than 2 lbs of set force applied
	Femur length does not shorten by more than 1-2 cm
User must be able to know force applied	Device shall measure applied force with accuracy of ± 2 lbs
	Force application information is available to user
Device shall be radiolucent for X-rays	Device must be made of components that do not obstruct the fracture during X-rays
The device must be compatible with use of most hospital beds and X-ray tables	Device fits within bounds of 36 x 80 x 24 inches

Proposed Solution (DEMO time!)

	User Need
1.	Device shall apply traction to femur fracture per current treatment requirements
2.	The device shall maintain the set length and position of femur throughout treatment
3.	User must be able to know forces applied
4.	Device shall be radiolucent for X-rays and CT scans
5.	The device must be compatible with use of most hospital bed, including X-ray and CT beds



Can it apply a pulling force on bone pin?

YES!



Can maintain traction?

YES!



Can the amount of applied force be measured?

YES!



Can it be used in scans?

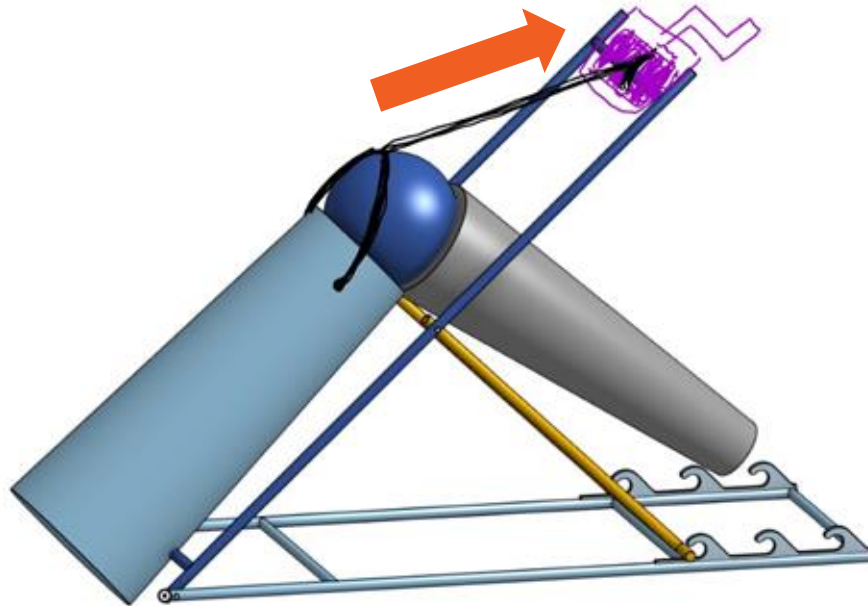
YES! (material)



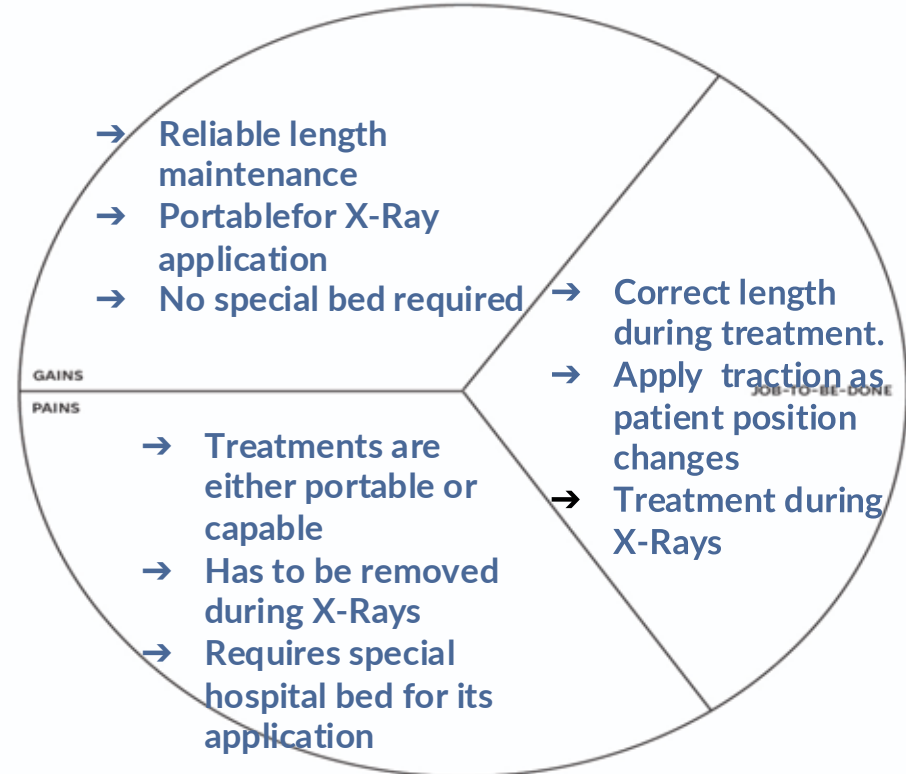
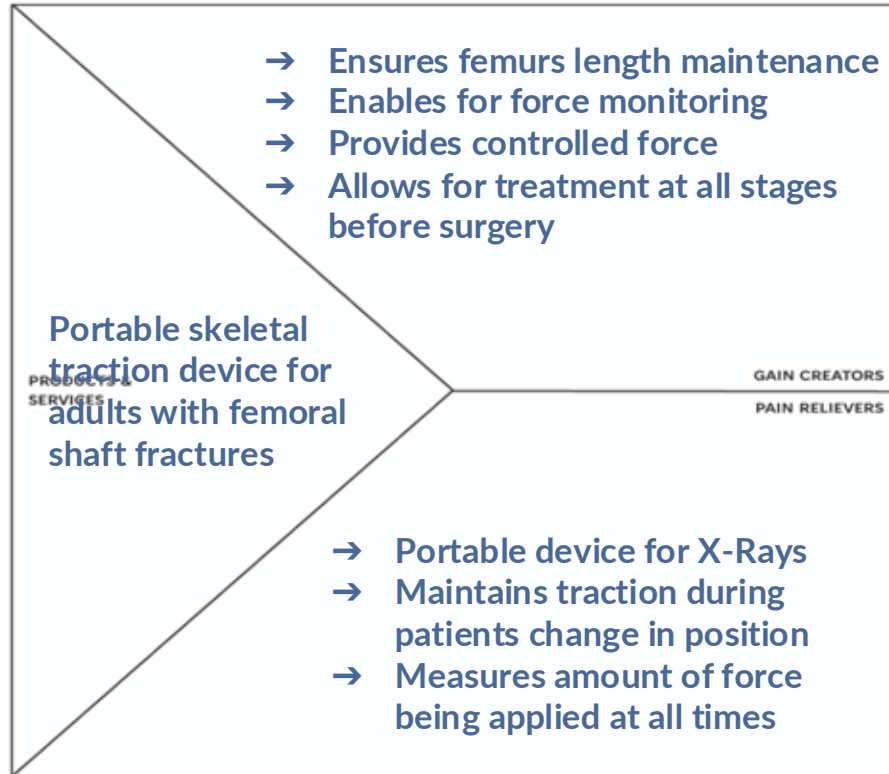
Can it be used with different beds?

YES!

Device Application



Value Proposition





Value Proposition

Our portable skeletal traction device allows orthopedic surgeons to apply and monitor traction, maintaining correct femur length even **during patient repositioning**.

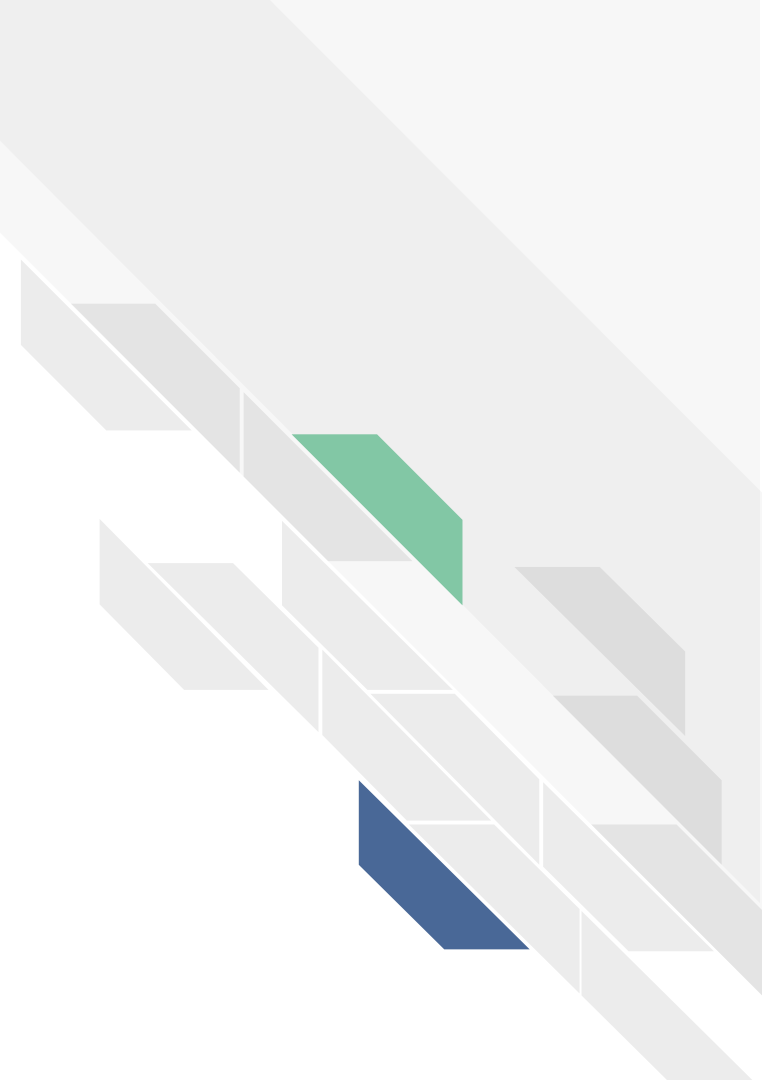
Unlike current methods, the device is **compatible with most standard hospital beds**, and continued **traction throughout scans** enables more accurate treatment plans for better outcomes.

Impact on Patient

- Faster treatment application
- Avoids under and excess traction complications
- Reduces risk of additional injury during patient transportation
- Increase patients comfortability
- Correct traction = Faster surgery
 - Lower risk for complications during surgery



CAD Modeling and Simulations



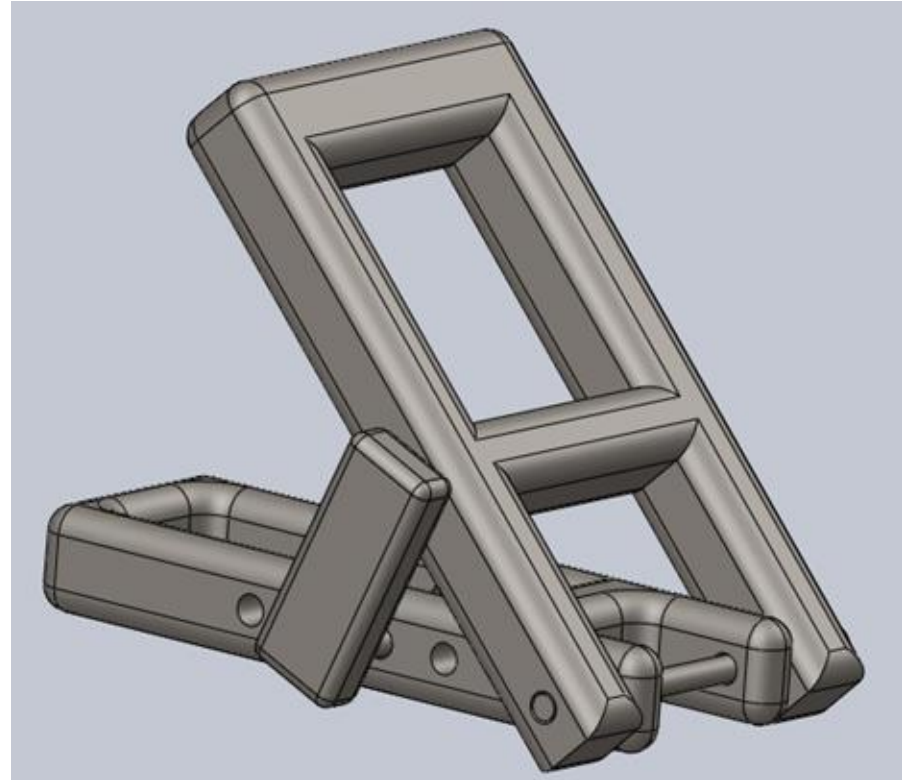
Solidworks Modeling Overview

Traction Solidworks Modeling

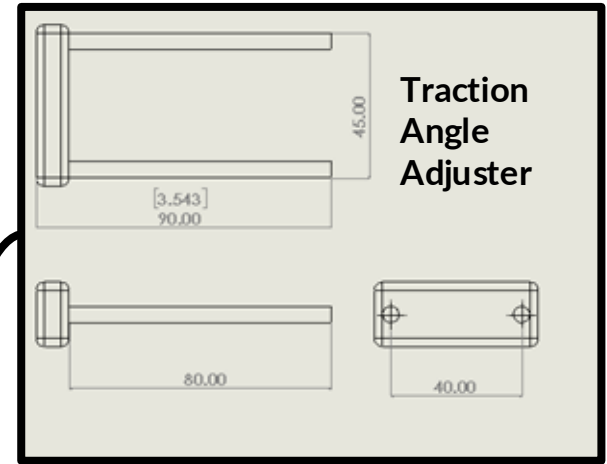
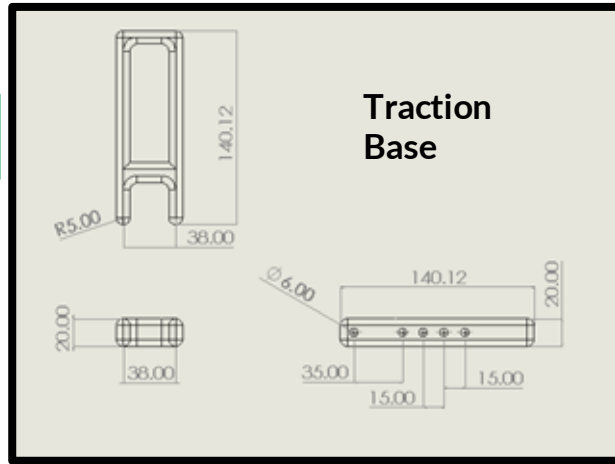
- Iterative process for prototyping fixtures
- Test different designs to identify ideal solution

Long Term Strategy

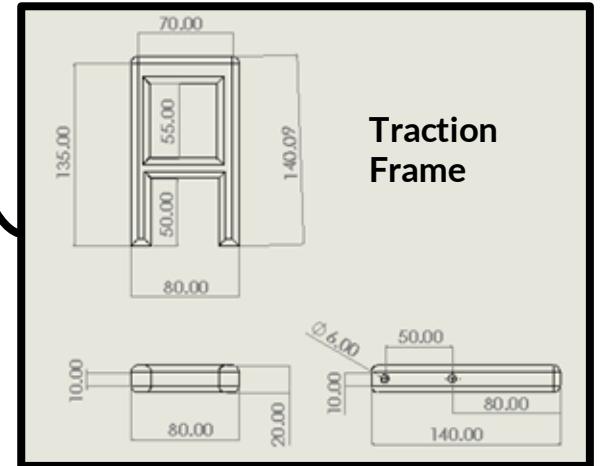
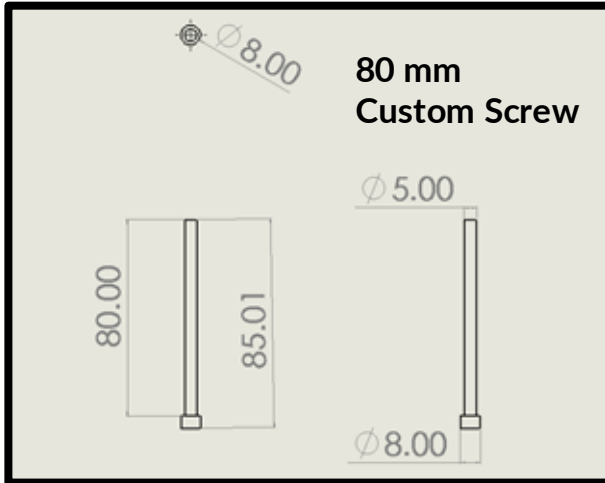
- Simulate various fixtures for improved performance
- Identify ideal material for assembly



First Iteration Prototype Components



Final Assembly



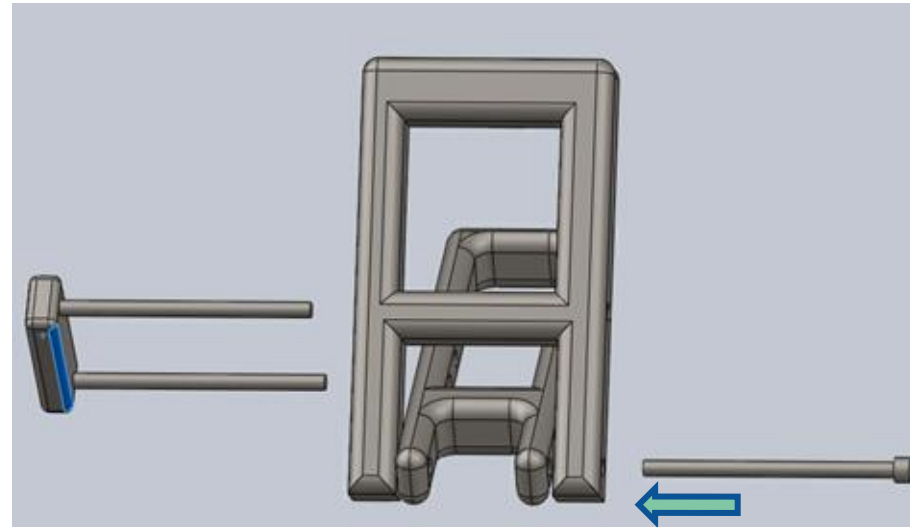
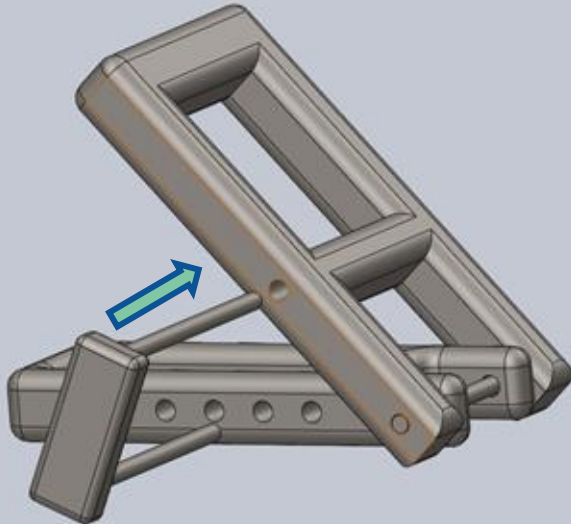
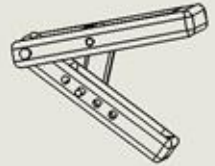
Initial Prototype Assembly

4 Component Assembly

1. Base slides under frame
2. Screw → intersection of base and frame
3. Angle adjuster → midsection of frame and base



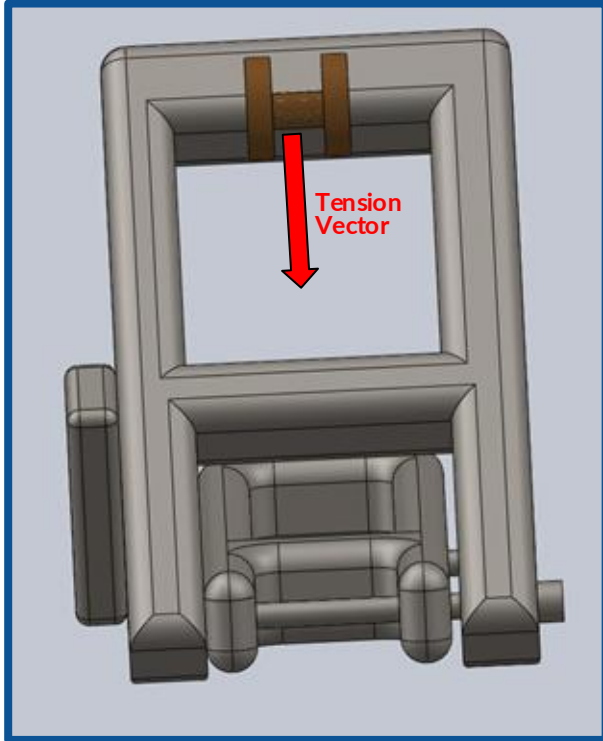
First Iteration
Traction Assembly



Key Features of Model

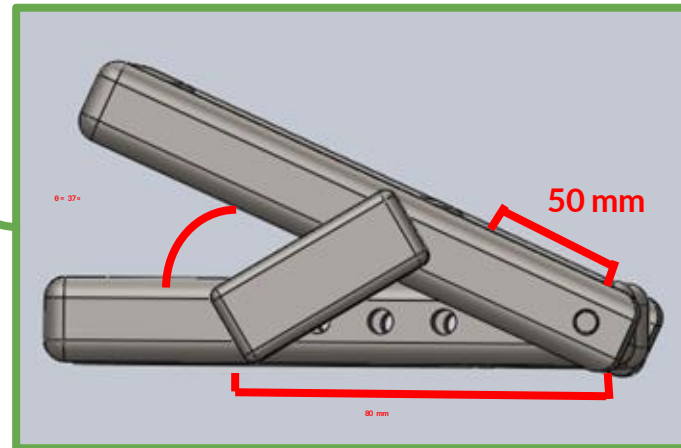
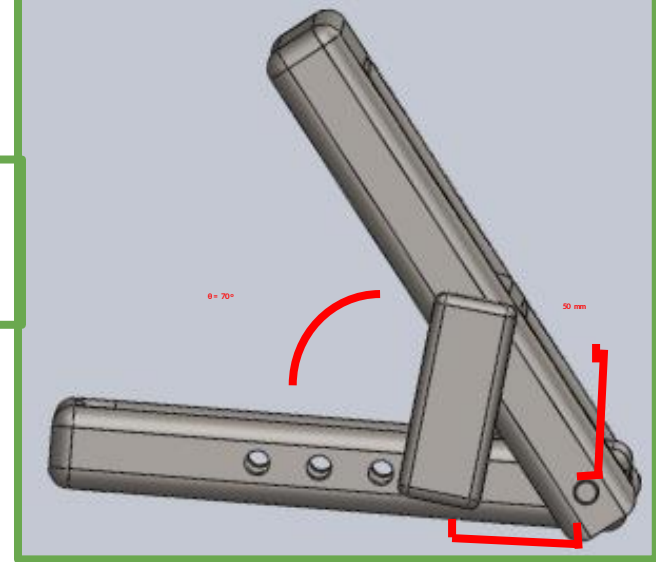
Core Feature 1

- Winch Mechanism



Core Feature 2

- Adjustable Angle



Material Considerations

Primary

- Strength and Fatigue
- Weight
- Imaging Compatibility
- Biocompatibility

Secondary

- Manufacturability
- Cost
- Corrosion Resistance
- Cleanability

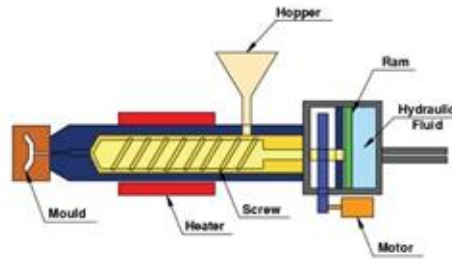


Autoclavable materials
→ sterilization



Low weight for user safety

Injection Molding



Material impacts manufacturing
(plastic = moldability)



X Ray compatible

Material options

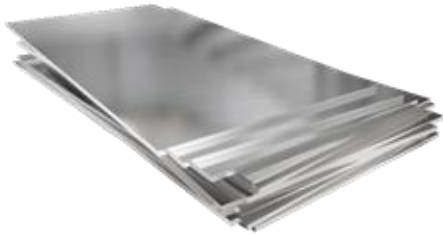
Delrin Thermoplastic



Aluminum Alloys (6061-T6)



Stainless Steel



Carbon Fiber Reinforced Polymer



Performance Simulations

Stress Test

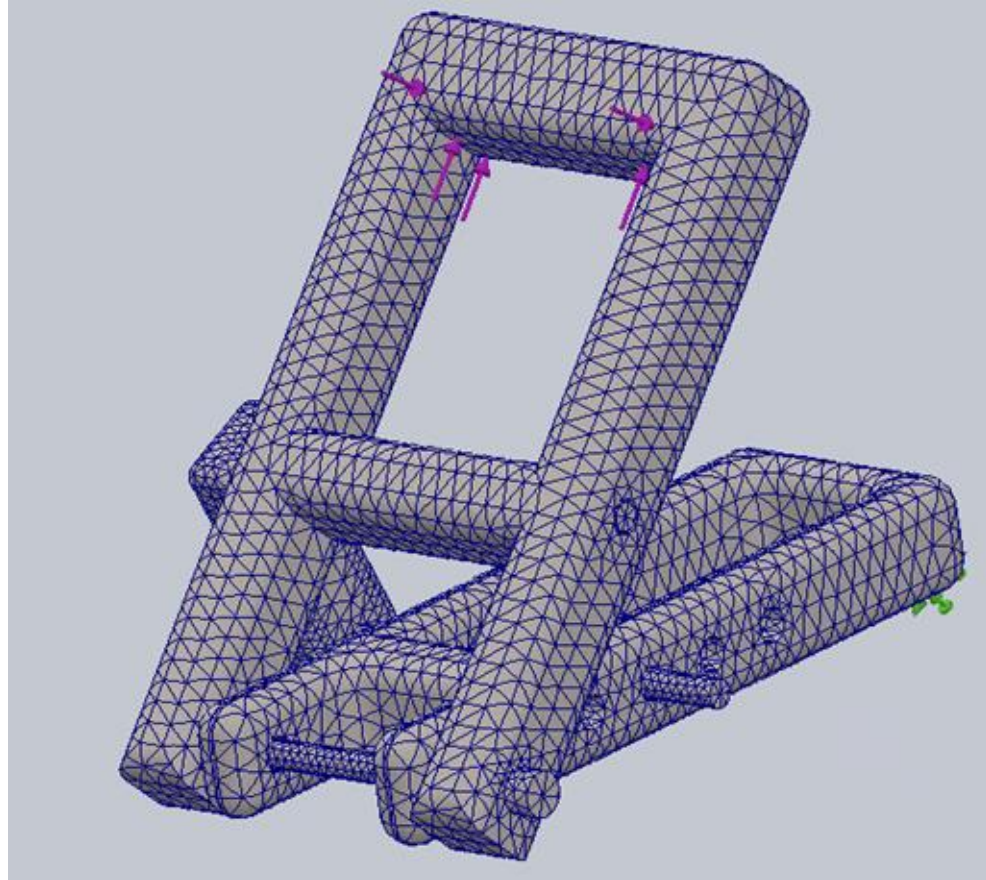
- Determine fixture fracture under load

Fatigue Analysis

- Assess load cycles to fracture

Result Analytics

- Redesign geometry to optimize performance
- Material selection informed by simulations



Mechanical Testing Sim - Stress Test

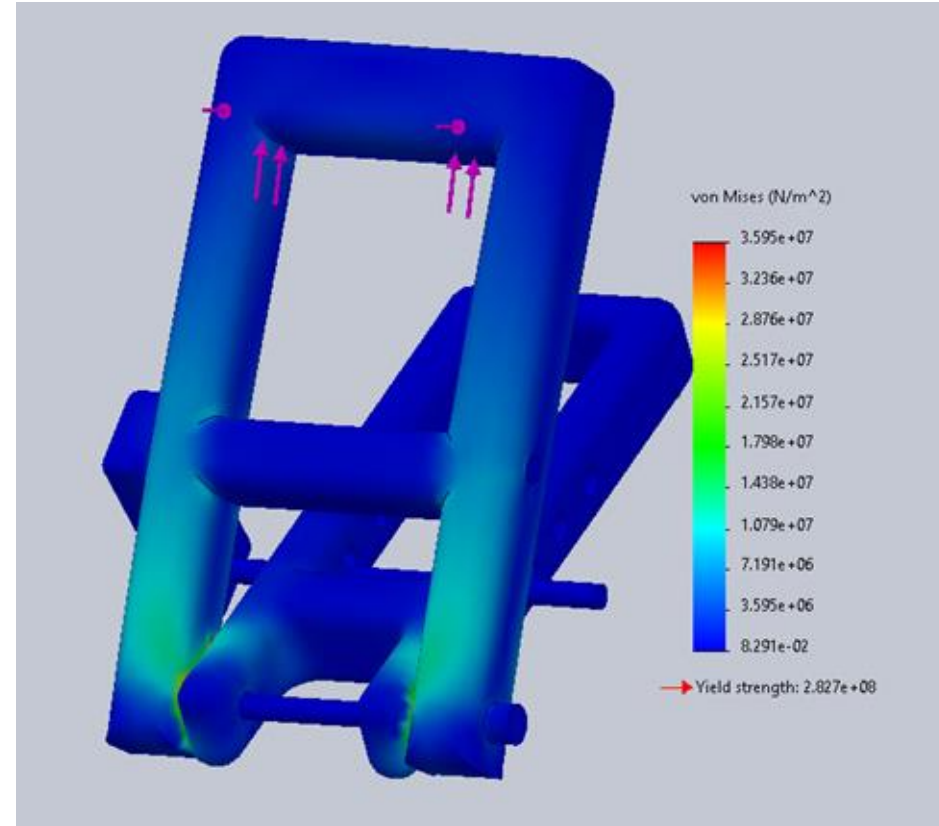


Stress Simulating

- Apply Material
- Apply Force → 250 N (56 lb)
 - Force determines max von mises value
- Fracture → Max Stress (Red) > Yield Strength

Property	Value	Units
Elastic Modulus	2.049999984e+11	N/m ²
Poisson's Ratio	0.29	N/A
Shear Modulus	7.999999987e+10	N/m ²
Mass Density	7858.000032	kg/m ³
Tensile Strength	425000003.2	N/m ²
Compressive Strength		N/m ²
Yield Strength	282685049	N/m ²

Material Property: 1023 Carbon Steel (seen on example)



Visualized stress distribution
(Max stress depends on design, agnostic to material)

Material Selection

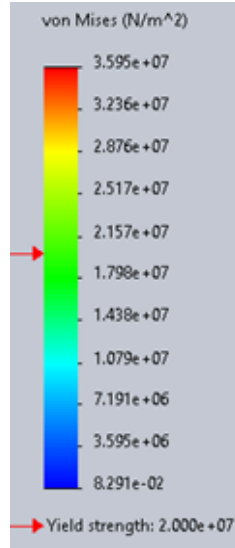
Summary

- Applied force design requirement
 - Greater max stress on device
- Device Geometry
 - Test device design
 - Minimize max stress
- Material
 - Select greater yield strength than max stress point
 - Compare stress to SN Curve

Fracture Results

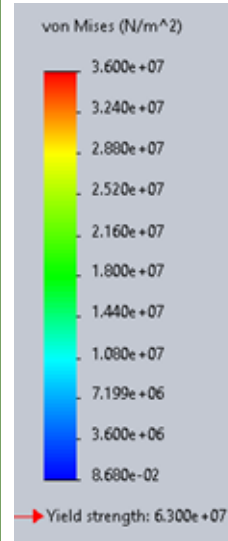


Balsa Wood



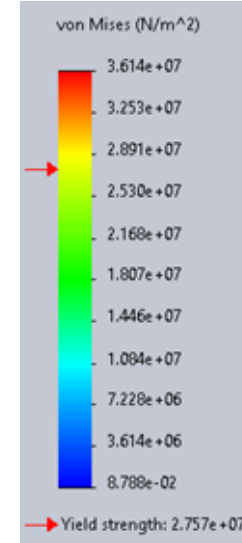
Fracture

Delrin (Thermoplastic)



No Fracture

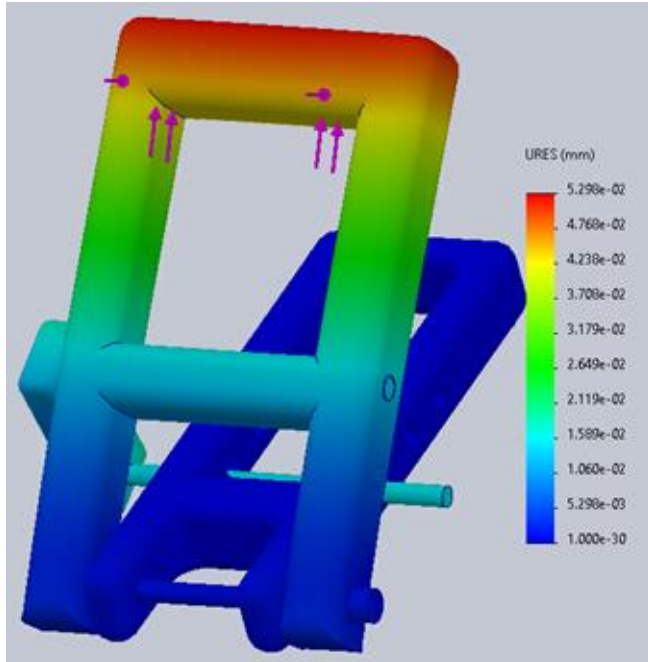
1060 Aluminum Alloy



Fracture

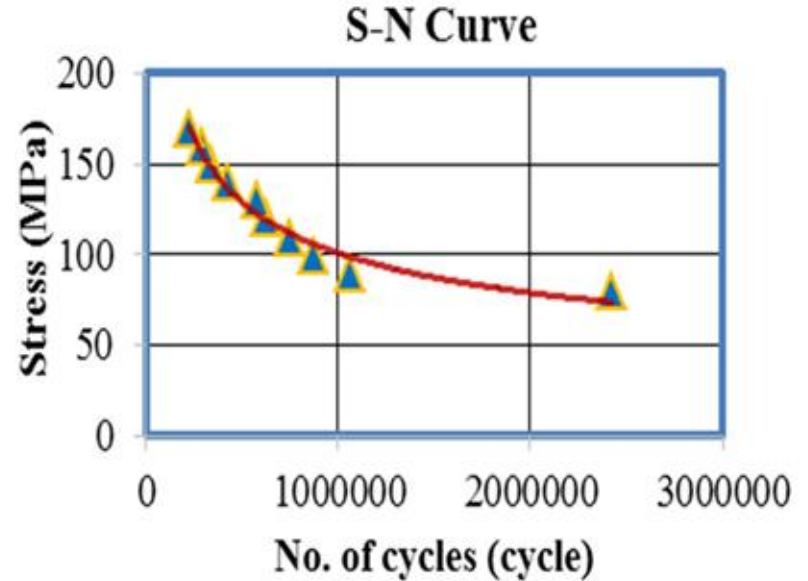
Additional Simulation Tests

Displacement Testing



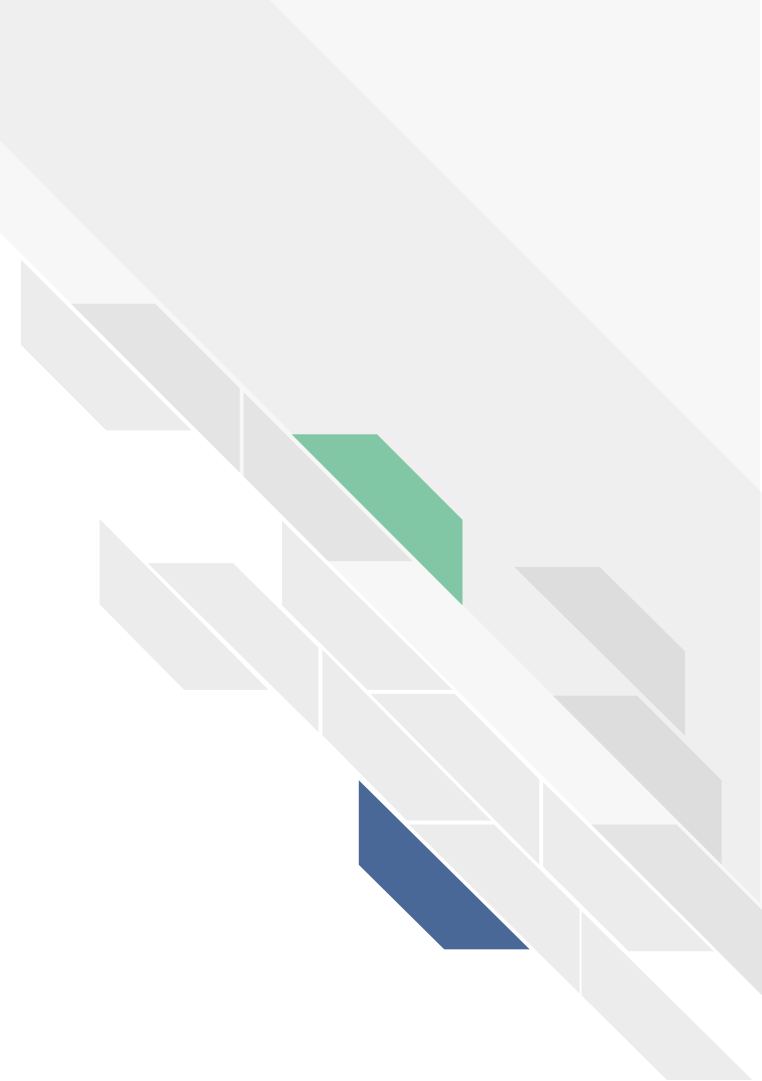
Observed bending at head of frame

Fatigue Testing



Fatigue curve for AISI 1008 Carbon Steel

Testing Prototyping and Manufacturing



Initial Testing

- Focus on tension mechanism
- Simulate prolonged traction application
- Check for correct force and length
- Evaluate frame for any wear/deformation
- Ensure leg stand-in remains in viable position



Shipping handling testing, Drop testing, Environmental testing (in chamber) (this might not be in our umbrella for presentation)

Prototyping

- Simulate device properties in SolidWorks
- Use cheap material to iterate through ideas and test functionality
- Use prototypes to inform design decisions for future/more advanced iterations



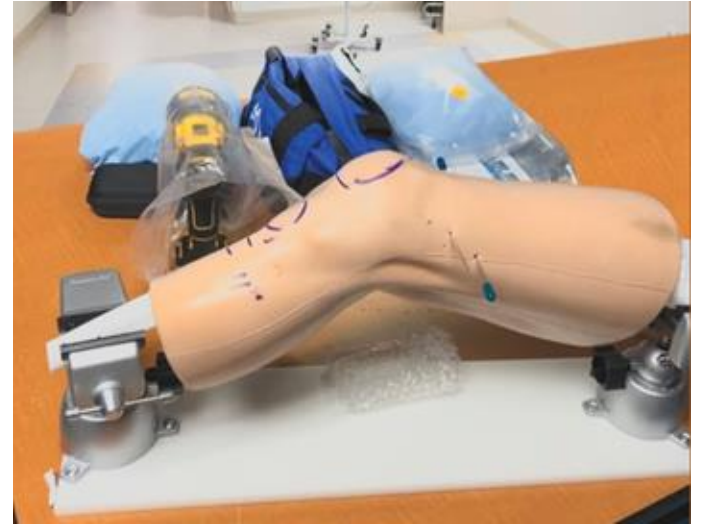
Scaled Up Production

- Metal Manufacturing
 - Forging
 - Milling/Shaping
 - Finishing
 - Extrusion
- Gas Sterilization
 - EtO or H₂O₂
- Packaging
 - Device frame collapses into large box
 - Pin provided in sterile, puncture resistant pouch



Resources

- Raw materials and components to construct prototypes and device
- EPIC/SILab access for machining
- Simulation apps





Anticipated challenges

- Application of tension in device at magnitudes of 40lb.
- Winch positioning
- Sizing of the device
- Balancing features vs. complexity
- Balancing strength and size/weight
- Device effectiveness if bumped





Timeline

Ordering PVC Frame (For Testing) -
THIS WEEK

Ordering Final Frame Components
Raw Materials → Aluminum tubing,
Bolts, Screws, Aluminum Sheets

Testing and ordering rope

Finalize a design
Finalize Sizing

Questions?

